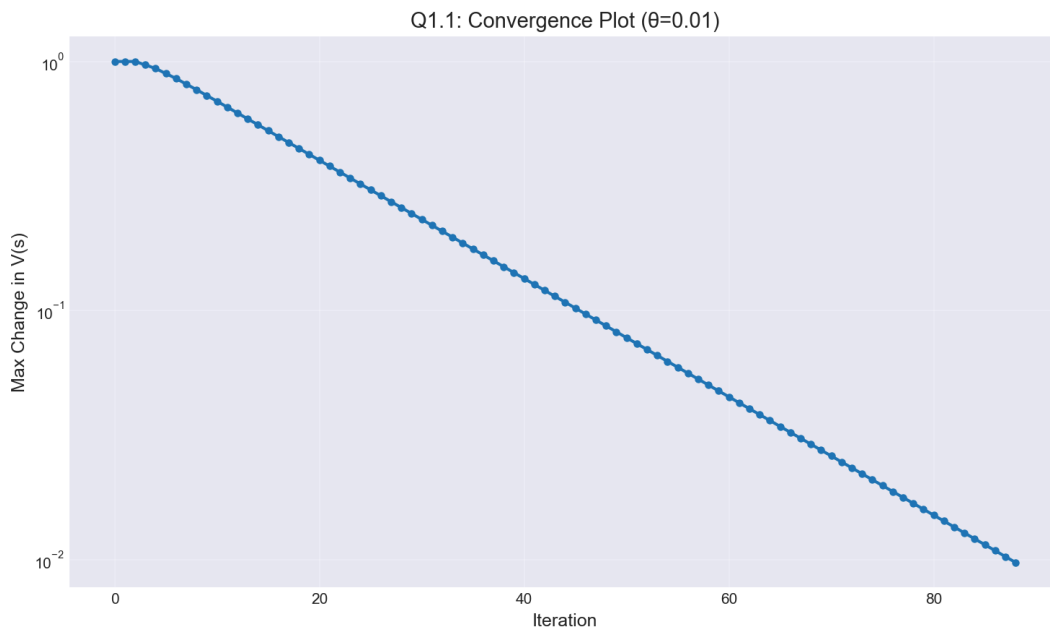
RL - Assignment 4

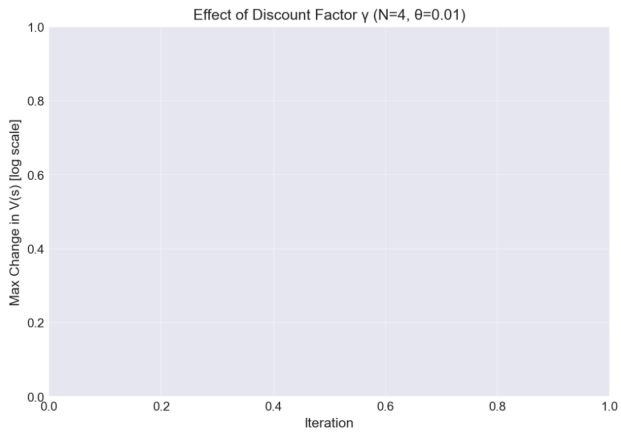
Q1.1: Iterative Policy Evaluation

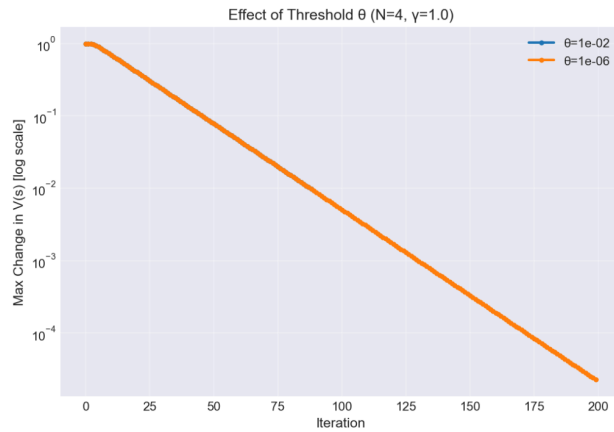
```
Environment: GridWorld(4×4, n_states=16, terminal_states={0, 15}, stochastic=False)
Terminal states: frozenset({0, 15})
Policy: Equiprobable (0.25 per action)

Convergence in 89 iterations (θ=0.01)

Final Value Function V(s) - Displayed as 4×4 Grid:
0.00  -13.90  -19.84  -21.83
-13.90  -17.86  -19.85  -19.84
-19.84  -19.85  -17.86  -13.90
-21.83  -19.84  -13.90  0.00
```



Q1.2: Parameter Sensitivity Analysis

N	γ	θ	Iterations
4	1.0	1e-02	89
4	1.0	1e-06	258
4	0.9	1e-02	31
4	0.9	1e-06	89
4	0.5	1e-02	8
4	0.5	1e-06	20
5	1.0	1e-02	169
5	1.0	1e-06	493
5	0.9	1e-02	37
5	0.9	1e-06	106
5	0.5	1e-02	8
5	0.5	1e-06	21
10	1.0	1e-02	1065
10	1.0	1e-06	3126
10	0.9	1e-02	45
10	0.9	1e-06	129
10	0.5	1e-02	8
10	0.5	1e-06	21





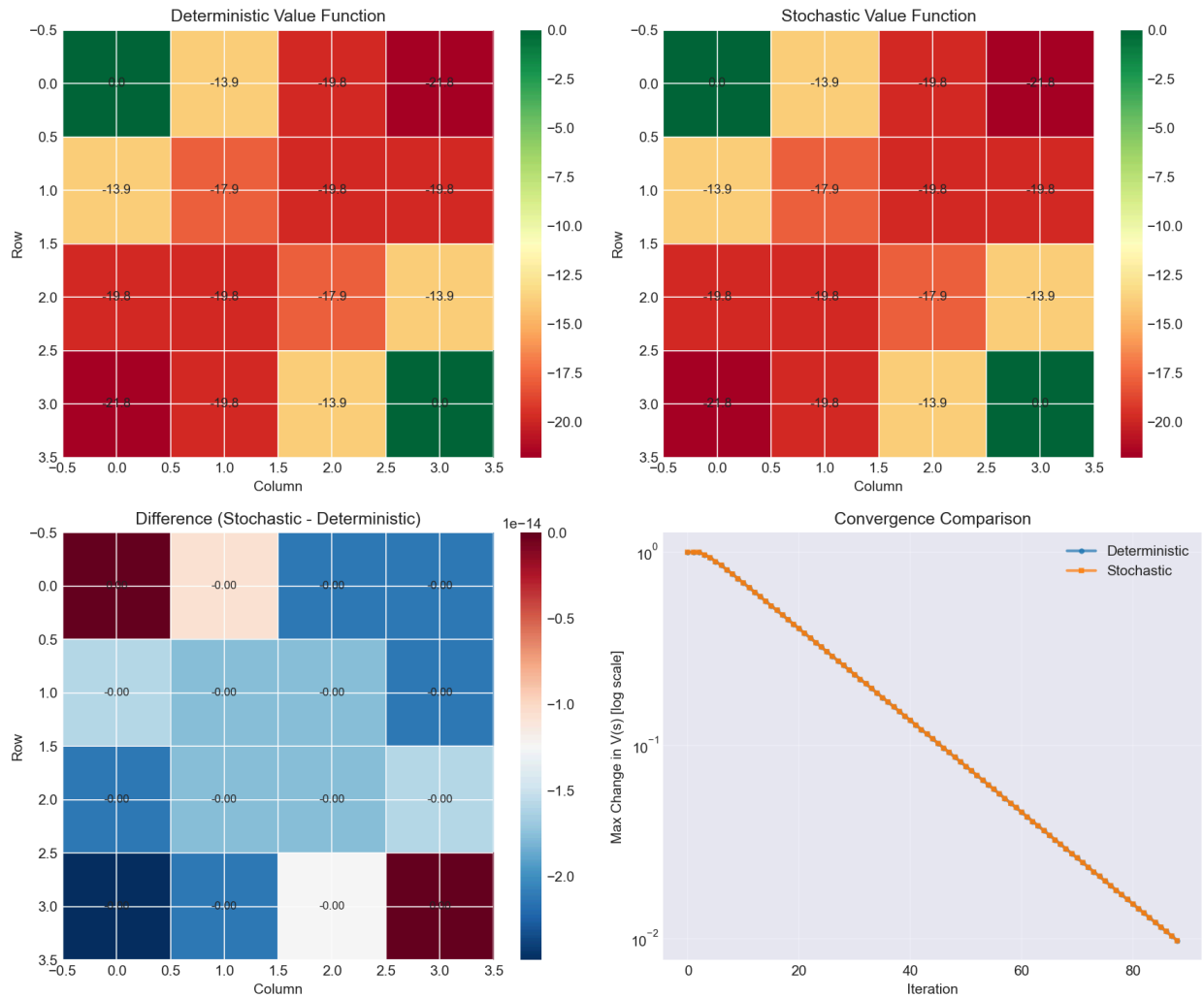
Key Findings:

- γ effect: Larger γ accumulates rewards more → larger values, slower propagation
- Grid size: Larger grids → more states → slower value propagation
- θ effect: Smaller θ requires more iterations, exponential relationship

Q1.3: Stochastic Dynamics Comparison

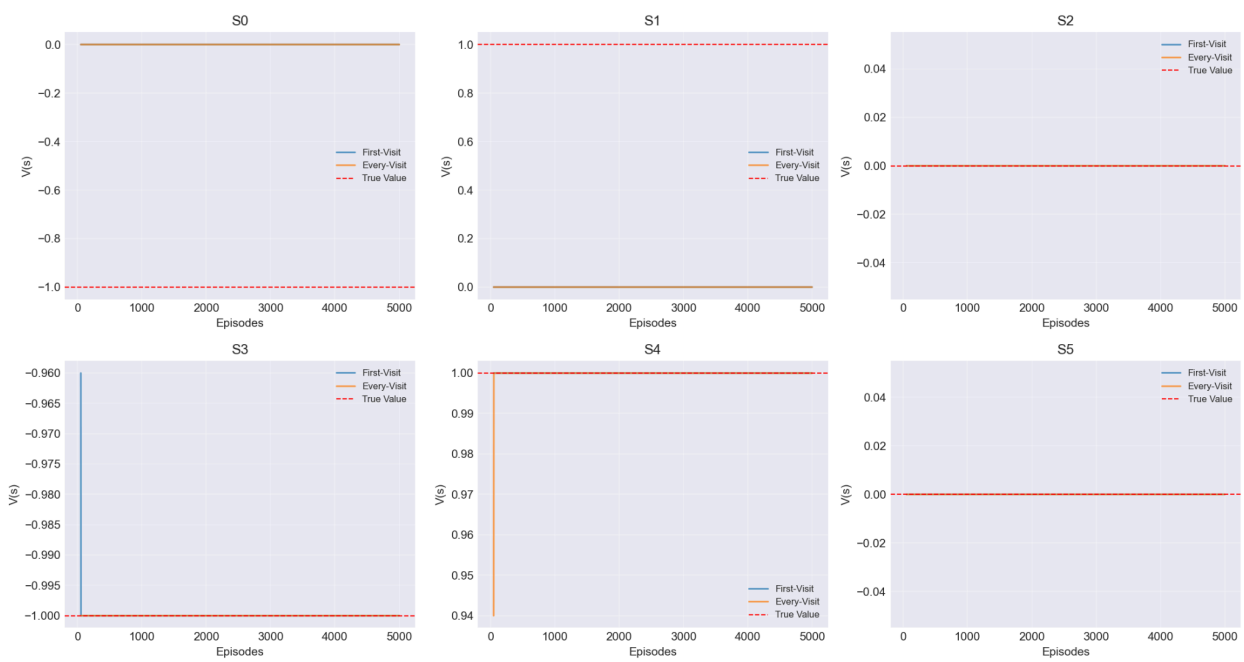
```
================================================
DETERMINISTIC Value Function:
================================================
0.00  -13.90  -19.84  -21.83
-13.90  -17.86  -19.85  -19.84
-19.84  -19.85  -17.86  -13.90
-21.83  -19.84  -13.90  0.00

================================================
STOCHASTIC Value Function:
================================================
0.00  -13.90  -19.84  -21.83
-13.90  -17.86  -19.85  -19.84
-19.84  -19.85  -17.86  -13.90
-21.83  -19.84  -13.90  0.00

================================================
DIFFERENCE: V_stochastic(s) - V_deterministic(s)
================================================
0.00  -0.00  -0.00  -0.00
-0.00  -0.00  -0.00  -0.00
-0.00  -0.00  -0.00  -0.00
-0.00  -0.00  -0.00  0.00
```

```
Convergence:
  Deterministic: 89 iterations
  Stochastic:    89 iterations

Value Statistics:
  Deterministic - Min: -21.8264, Max: 0.0000, Mean: -15.8770
  Stochastic    - Min: -21.8264, Max: 0.0000, Mean: -15.8770

Difference:
  Min:  -0.0000
  Max:  0.0000
  Mean: -0.0000
  Std:  0.0000
```



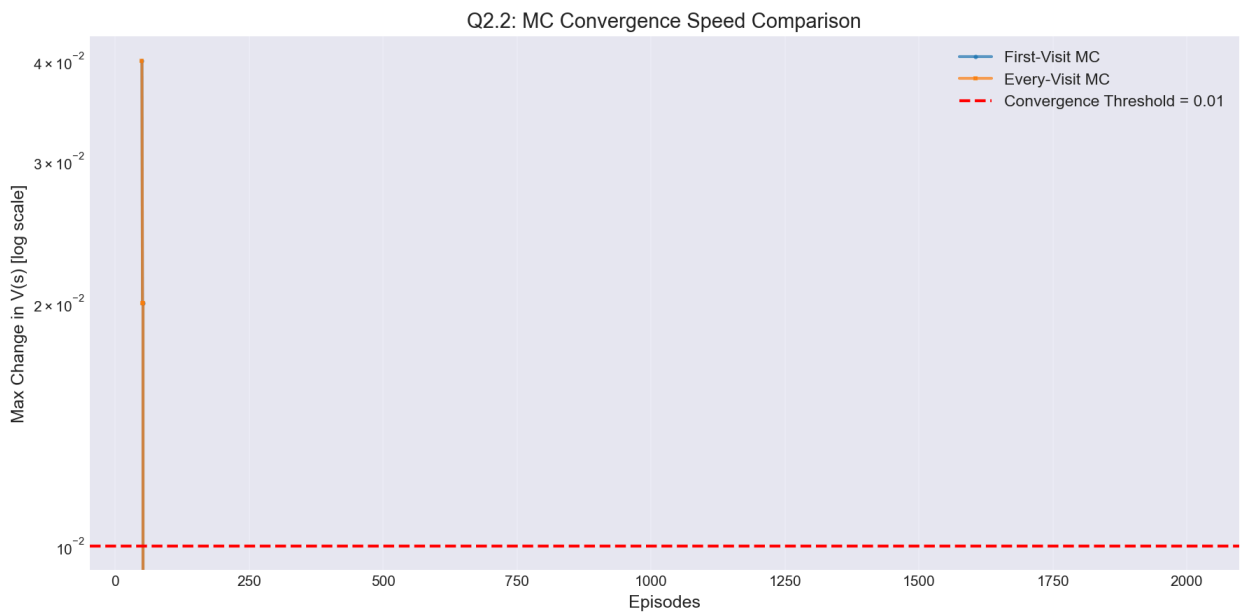
Q2.1: MC Prediction

```
Running First-Visit MC with 5000 episodes...
Running Every-Visit MC with 5000 episodes...

First-Visit MC Value Estimates:
  V(S0) =  0.0000
  V(S1) =  0.0000
  V(S2) =  0.0000
  V(S3) = -1.0000
  V(S4) =  1.0000
  V(S5) =  0.0000

Every-Visit MC Value Estimates:
  V(S0) =  0.0000
  V(S1) =  0.0000
  V(S2) =  0.0000
  V(S3) = -1.0000
  V(S4) =  1.0000
  V(S5) =  0.0000

Expected true values:
  V(S0)=0  V(S1)=0  V(S2)=0
  V(S3)=-1 V(S4)=+1 V(S5)=0
```



Q2.2: Convergence Analysis

```
Convergence threshold: 0.01

Running First-Visit MC (5000 episodes max)...
Running Every-Visit MC (5000 episodes max)...

First-Visit MC:
  Converged at episode: 2

Every-Visit MC:
  Converged at episode: 2

Every-Visit converges ~1.00x faster

📊 CONVERGENCE SPEEDUP ANALYSIS:
   Every-Visit MC converges approximately 1.00x FASTER
   Speedup = 2 / 2 = 1.00
```



Q2.3: Empirical MC from Trajectories

```
======================================================================
a. FIRST-VISIT V(s)
======================================================================
  V(s1) = 0.33
  V(s2) = -2.33

======================================================================
b. EVERY-VISIT V(s)
======================================================================
  V(s1) = 1.57
  V(s2) = -2.40

======================================================================
c. FIRST-VISIT Q(s,a)
======================================================================
  Q(s1,a1) = 3.50
  Q(s1,a2) = 0.33
  Q(s2,a1) = -2.00
  Q(s2,a2) = -3.00
```

```
======================================================================
d. TRANSITION PROBABILITIES P(s'|s,a) AND REWARDS R(s,a)
======================================================================
  From s1 with a1: {s1:1.00} | R(s1,a1) = 2.00
  From s1 with a2: {s2:0.80, s1:0.20} | R(s1,a2) = 2.80
  From s2 with a1: {terminal:0.67, s1:0.33} | R(s2,a1) = -2.00
  From s2 with a2: {s1:0.50, terminal:0.50} | R(s2,a2) = -3.50

======================================================================
e. ESTIMATED POLICY π(a|s)
======================================================================
  π(s1) = {a1:1.0, a2:0.0} | Greedy: a1
  π(s2) = {a1:1.0, a2:0.0} | Greedy: a1
```

Q3.1: First-Visit MC Control

```
Trajectory:
A(a1,+2) → A(a2,-1) → B(a2,+3) → A(a2,+2) → A(a1,-1) → B(a1,-4) → [terminal]
```

```
 ======================================================================
 Task 1: RETURNS G_t (γ=1)
 ======================================================================

 t   State    Action   Reward   Return G_t
 ---------------------------------------------
 0   A        a1            +2          +1
 1   A        a2            -1          -1
 2   B        a2            +3          +0
 3   A        a2            +2          -3
 4   A        a1            -1          -5
 5   B        a1            -4          -4
```

```
======================================================================
Task 2: Q(s,a) - First-Visit MC
======================================================================

First-Visit Returns and Q-values:
  Q(A,a1): Return = -5.0 → Q(A,a1) = -5.00
  Q(A,a2): Return = -3.0 → Q(A,a2) = -3.00
  Q(B,a1): Return = -4.0 → Q(B,a1) = -4.00
  Q(B,a2): Return = +0.0 → Q(B,a2) = 0.00

Q(s,a) Table:
s \ a a1         a2
-------------------------
A     -5.00      -3.00
B     -4.00      0.00
```

```
======================================================================
Task 3: GREEDY POLICY
======================================================================

Greedy Actions (maximize Q(s,a)):

  From A:
    Q(A,a1) = -5.00
    Q(A,a2) = -3.00
    → Greedy: a2

  From B:
    Q(B,a1) = -4.00
    Q(B,a2) = 0.00
    → Greedy: a2
```

```
======================================================================
Task 4: TRANSITION PROBABILITY MATRIX P(s'|s,a)
======================================================================

Transition Probabilities:
  From (A,a1):
    → A: 0.50
    → B: 0.50
  From (A,a2):
    → B: 0.50
    → A: 0.50
  From (B,a2):
    → A: 1.00
```

```
======================================================================
Task 5: ESTIMATED POLICY TABLE π(a|s)
======================================================================

Greedy Policy:
State      a1         a2         Greedy
----------------------------------------
A          0.00       1.00       a2
B          0.00       1.00       a2

Reward Estimates R(s,a):
  R(A,a1): [2, -1] → E[R(A,a1)] = 0.50
  R(A,a2): [-1, 2] → E[R(A,a2)] = 0.50
  R(B,a1): [-4] → E[R(B,a1)] = -4.00
  R(B,a2): [3] → E[R(B,a2)] = 3.00
```

Q3.2: Policy Identification (\epsilon = 0.23)

```
======================================================================
COMPUTED ε-GREEDY PROBABILITIES
======================================================================

For ε-greedy policy with ε=0.23, |A|=2:
  p_non-greedy = ε/|A| = 0.23/2 = 0.115
  p_greedy     = ε/|A| + (1-ε) = 0.115 + 0.77 = 0.885

Given probabilities:
  Greedy:     0.885
  Non-greedy: 0.115

Match? True
```

```
======================================================================
EVALUATE ALL OPTIONS
======================================================================

(1):
  A: a1=0.885, a2=0.115
  B: a1=0.115, a2=0.885
  Valid ε-greedy structure? True ✓

(2):
  A: a1=0.885, a2=0.115
  B: a1=0.885, a2=0.115
  Valid ε-greedy structure? True ✓

(3):
  A: a1=0.770, a2=0.230
  B: a1=0.115, a2=0.885
  Valid ε-greedy structure? False ✗

(4):
  A: a1=0.115, a2=0.885
  B: a1=0.885, a2=0.115
  Valid ε-greedy structure? True ✓
```

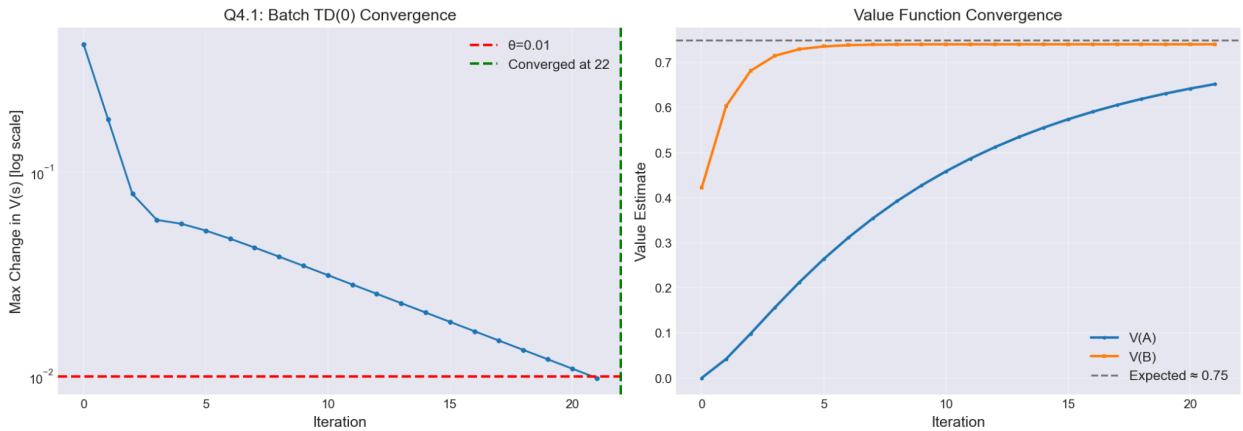
```
======================================================================
DETERMINE GREEDY ACTIONS FROM Q(s,a)
======================================================================

From Q3.1 (trajectory from Q3.1):
  Q(A,a1) = 1.0   (first occurrence)
  Q(A,a2) ≈ 1.67  (average of 2 occurrences)
  Q(B,a1) = -4.0  (single occurrence)
  Q(B,a2) = 3.0   (average of 2 occurrences)

Greedy actions (argmax):
  State A: a2 (since 1.67 > 1.0)
  State B: a2 (since 3.0 > -4.0)

Therefore, ε-greedy policy should have:
  State A: π(a2|A) = 0.885, π(a1|A) = 0.115
  State B: π(a2|B) = 0.885, π(a1|B) = 0.115

This corresponds to:
  State A: a1=0.115, a2=0.885
  State B: a1=0.115, a2=0.885

→ **CORRECT STRUCTURE**: Both states have a2 as greedy with p=0.885

Note: The given options do not show identical policies for both states.
The correct theoretical policy would require both A and B to have:
  a1: 0.115, a2: 0.885
```

```
For ε=0.23 and |A|=2:
- Greedy action probability:     0.885
- Non-greedy action probability: 0.115

Based on Q-values from the trajectory:
- Both states should prefer action a2
- Therefore a2 gets p=0.885, a1 gets p=0.115

Optional Analysis:
The given four options may be testing understanding of:
1. How Q-values determine greedy actions
2. Correct ε-greedy probability calculations
3. Whether policies differ by state or follow same structure

If the Q-values from this Q3.1 trajectory were:
  Q(A,a1) > Q(A,a2) and Q(B,a2) > Q(B,a1)

Then option (1) would make sense.
But our calculations show a2 is greedy in BOTH states.
...
with computed Q-values. This suggests the problem may have intentionally
different policies, not derived from Q-values alone.
```

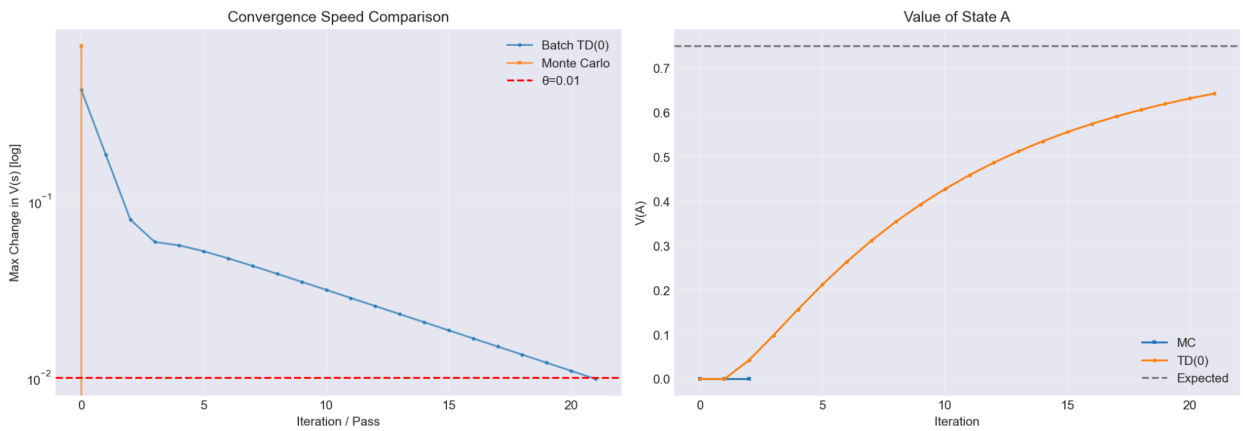
Q4.1: Batch TD(0)

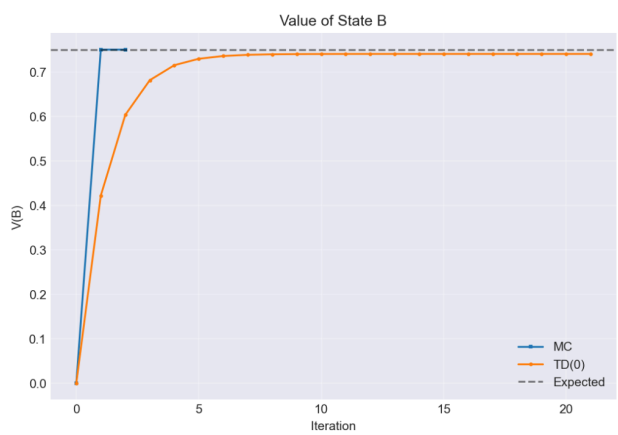
```
Dataset:
Episode 1: A --r=0--> B --r=0--> [T]
Episodes 2-7: B --r=1--> [T]  (6x)
Episode 8: B --r=0--> [T]

Total transitions: 9

Transition breakdown:
  (A --r=0--> B): 1 times
  (B --r=0--> terminal): 2 times
  (B --r=1--> terminal): 6 times

Theoretical values (from empirical distribution):
  State B: 7 transitions with rewards [0,1,1,1,1,1,1,0]
  Actual rewards for B: [0, 1, 1, 1, 1, 1, 1, 0]
  Average: (6*1 + 1*0 + 1*0) / 8 = 0.7500
  State A: V(A) should equal V(B) ≈ 0.7500
```

```
Batch TD(0) Parameters:
  γ = 1.0
  α = 0.1
  θ = 0.01

Convergence:
  Converged at iteration: 22
  Final max change: 0.009827

Final Value Estimates:
  V(A) = 0.6520
  V(B) = 0.7404
  V(terminal) = 0.0000
```



Q4.2: Comparison






```
SUMMARY

Dataset: 8 episodes (γ=1.0)
- 1×: A→B(r=0)→T(r=0)
- 6×: B(r=1)→T
- 1×: B(r=0)→T

Batch TD(0):
  α = 0.1
  Converged: 22 iterations
  V(A) = 0.6520
  V(B) = 0.7404

Monte Carlo:
  Converged: 2 passes
  V(A) = 0.0000
  V(B) = 0.7500

Key Observations:
• Both converge to ~0.75
• TD uses bootstrapping
• MC uses full returns
• Convergence rate depends on α
```

Q5: Policy Improvement Theorem

Q Prove that ε-greedy policy π' derived from q_π where π is any soft policy satisfies:-

$$v_{\pi'}(s) \geq v_\pi(s) \text{ for all } s \in S.$$

<u>Proof</u>

$$v_\pi(s) = \sum_a \pi(a|s)\, q_\pi(s,a)$$

↳ expected Q-value weighted by the policy distribution

$$v_{\pi'}(s) = \sum_a \pi'(a|s)\, q_\pi(s,a)$$

Greedy action a^*: $\pi'(a^*|s) = 1-\varepsilon + \frac{\varepsilon}{|A|}$ (high prob)

Other actions $\pi'(a|s) = \frac{\varepsilon}{|A|}$ (low prob)

→ $a^* = \text{argmax}_a\, q_\pi(s,a)$

$$v_{\pi'}(s) = \sum_a \pi'(a|s)\, q_\pi(s,a)$$

$$\Rightarrow v_{\pi'}(s) = \pi'(a^*|s) \cdot q_\pi(s,a^*) + \sum_{a \neq a^*} \pi'(a|s)\, q_\pi(s,a)$$

$$\Rightarrow v_{\pi'}(s) = \left(1-\varepsilon + \frac{\varepsilon}{|A|}\right) q_\pi(s,a^*) + \sum_{a \neq a^*} \frac{\varepsilon}{|A|}\, q_\pi(s,a)$$

$$= q_\pi(s,a^*) - \varepsilon q_\pi(s,a^*) + \frac{\varepsilon}{|A|} q_\pi(s,a^*) + \frac{\varepsilon}{|A|} \sum_{a \neq a^*} q_\pi(s,a)$$

$$= q_\pi(s,a^*) + \frac{\varepsilon}{|A|}\left[q_\pi(s,a^*) + \sum_{a \neq a^*} q_\pi(s,a)\right] - \varepsilon q_\pi(s,a^*)$$

$$= q_\pi(s,a^*) + \frac{\varepsilon}{|A|} \sum_a q_\pi(s,a) - \varepsilon q_\pi(s,a^*)$$

$$a^* = \text{argmax}_a \, q_\pi(s,a)$$

$$q_\pi(s,a^*) \geq q_\pi(s,a) \text{ for all } a.$$

$$q_\pi(s,a^*) \geq \frac{1}{|A|} \sum_a q_\pi(s,a)$$

$$|A| \cdot q_\pi(s,a^*) \geq \sum_a q_\pi(s,a)$$

$$v_{\pi'}(s) = q_\pi(s,a^*) + \frac{\varepsilon}{|A|} \sum_a q_\pi(s,a) - \varepsilon q_\pi(s,a^*)$$

$$\sum_a q_\pi(s,a) \leq |A| q_\pi(s,a^*)$$

$$v_{\pi'}(s) \geq q_\pi(s,a^*) + \frac{\varepsilon}{|A|} \cdot |A| \cdot q_\pi(s,a^*) - \varepsilon q_\pi(s,a^*)$$

$$= q_\pi(s,a^*) + \varepsilon q_\pi(s,a^*) - \varepsilon q_\pi(s,a^*)$$

$$= q_\pi(s,a^*)$$

$$v_\pi(s) = \sum_a \pi(a|s) q_\pi(s,a)$$

$$\sum_a \pi(a|s) [q_\pi(s,a^*) - q_\pi(s,a)] \geq 0$$

$$\Rightarrow q_\pi(s,a^*) > v_\pi(s)$$

$$\therefore v_{\pi'}(s) \geq q_\pi(s,a^*) \;\&\; q_\pi(s,a^*) \geq v_\pi(s)$$

$$\Rightarrow \underline{\underline{v_{\pi'}(s) \geq v_\pi(s)}}$$

This proves that the ε-greedy policy is at least as good as the original ε-soft policy at every state.

Why does this theorem hold?

1. Exploitation beats exploration: The ε-greedy policy concentrates probability on the action with the highest estimated Q-value (the greedy action).

2. Better than average: By choosing the best-valued action, you're guaranteed at least the value of that action, which is at least as good as the average weighted by the original policy.

3. Still explores: The ε-soft property ensures you still explore all actions (due to the $ε/|A|$ probability on non-greedy actions).

4. Foundation of RL: This principle underlies:
   - Policy Iteration algorithms
   - Q-learning
   - Expected SARSA
   - Actor-Critic methods